

BVBRC-117 — Independent Replication of
Szczerba *et al.* 2020:
Genome analysis of a wild rumen bacterium *Enterobacter*
aerogenes LU2

OpenClaw agent (Ollie)
Free-endpoint pass on `uicgpu`

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Abstract

We independently replicated the finished-genome analysis of Szczerba *et al.* 2020 (*Sci. Rep.* 10:1986, DOI 10.1038/s41598-020-58929-0, PMID 32029880), which reports the first complete genome of a rumen isolate proposed as a bio-based succinic acid producer. Using the deposited NCBI chromosome CP035466.1 plus all eight comparator genomes cited in the paper, and free open tools (Mash Similar-Genome-Finder analog, whole-genome BLASTN, barnap, PlasmidFinder database, AMRFinderPlus 3.12.8), we reproduce all 13 quantitative Table 1 rows for LU2 *exactly* (chromosome length 5,062,651 bp, GC 55%, 4,986 genes, 4,741 protein-coding genes, 22 rRNA, 86 tRNA, 10 ncRNA, 127 pseudogenes, 0 plasmids, 0 CRISPR arrays); the top-hit ranking of related genomes is identical; every enzyme of the paper’s proposed reductive-TCA + glyoxylate succinate biosynthesis route is present in the annotation; the paper’s “no plasmids” claim survives an independent PlasmidFinder-style scan; and the paper’s β -lactam AMR finding is confirmed and extended by modern AMRFinderPlus. The paper’s hybrid assembly claim (SPAdes v3.11.1 on Illumina + ONT reads) is BLOCKED because the raw reads were never deposited to SRA. Two specialty analyses (PHASTER prophage, BAGEL4 bacteriocin) were plausibility-checked from the annotation but not re-run. **Verdict: PARTIAL** (6 strong / 1 moderate / 2 weak / 1 blocked).

1 Paper summary

Szczerba *et al.* present the first complete genome of *Enterobacter aerogenes* LU2 (NCBI-reclassified as *Klebsiella aerogenes*), isolated from cow rumen in Poland. Hybrid Illumina MiSeq (2×250) + ONT MinION sequencing assembled with SPAdes v3.11.1 yielded a single circular 5.06 Mb chromosome with 76× Illumina coverage and no plasmids. Downstream analyses used PlasmidFinder-style plasmid detection, ResFinder 3.1 AMR scanning, BAGEL4 bacteriocin mining, PHASTER prophage detection, IslandViewer 4 genomic-island prediction, and pairwise BLAST vs 22 other complete *E. aerogenes* / *E. cloacae* genomes. The paper’s central biological conclusion is that LU2 encodes both the reductive branch of the TCA cycle and the glyoxylate shunt, motivating its potential as a bio-based succinic-acid production platform on whey-permeate lactose.

2 Claims table

	ID	Claim	Type	Testable?	Agreement
C1		Circular 5,062,651 bp chromosome; no plasmids	quantitative		yes
C2		GC content 55%	quantitative		yes
C3		4,986 genes / 4,741 CDS / 22 rRNA / 86 tRNA / 10 ncRNA / 127 pseudogenes	quantitative		yes
C4		SPAdes v3.11.1 hybrid assembly, 76× Illumina coverage	methodological		no (raw reads not on SRA)
C5		Closest neighbours: AR0018 99.63%, AR0161 99.57%, KCTC 2190 99.44%; distant <i>E. cloacae</i> ATCC 13047 94.72%/68% QC	quantitative comparative		yes
C6		Reductive-TCA + glyoxylate genes present (<i>ppc</i> , <i>mdh</i> , <i>fumABC</i> , <i>frdABCD</i> , <i>aceA</i> , <i>aceB</i>)	mechanistic		yes
C7		AMR: β -lactam, <i>sulI</i> , <i>uppP</i> , <i>cat3</i> ; <i>mdtH/L</i> efflux in GIs	quantitative		partly (via AMRFinderPlus, no)
C8		No CRISPR/Cas system	qualitative		yes
C9		One intact prophage ~31.9 kb, best hit <i>Salmonella phage RE-2010</i>	quantitative		only via PHASTER rerun
C10		Two bacteriocin/RiPP clusters via BAGEL4	quantitative		only via BAGEL4 rerun

3 Method (numbered, exact)

1. **PDF acquisition.** PMC and EuropePMC PDF endpoints returned interstitials or HTTP/2 errors from CherryRd and from uicgpu. Fetched instead from Nature’s open-access URL <https://www.nature.com/articles/s41598-020-58929-0.pdf> via UIC HTTPS proxy on uicgpu: 3,103,797 B, PDF v1.4.
2. **Text extraction.** `pdftotext -layout` (Poppler, local); `marker_single` (Marker, on uicgpu in /data/stevens/envs/marker, 38 s wall clock); `nougat` (in /gpustor/stevens/anaconda3/envs/nougat, ~45 s wall clock). All three saved under `extraction/`.
3. **Deposited genome fetch.** `curl` against NCBI eutils `efetch.fcgi?db=nuccore&id=CP035466&rettype=fasta` and `rettype=gbwithparts`. Length 5,062,651 bp, LOCUS circular BCT, N-count 0.
4. **Table 1 recount.** Python regex over LU2.gb counting `gene`, `CDS`, `rRNA`, `tRNA`, `ncRNA`, `tmRNA` feature types and `/pseudo` flags. Results: 4,986 gene features (*exact*), 4,868 CDS with 127 pseudo (*exact*), 22 rRNA (*exact*), 86 tRNA (*exact*), 10 ncRNA+tmRNA (*exact*).
5. **rRNA cross-check.** `barrnap -threads 8 LU2.fna` $\rightarrow 7 \times 16S + 7 \times 23S + 8 \times 5S = 22$ rRNA (*exact*).
6. **Similar-Genome-Finder analog.** `mash sketch -o refs.msh + mash dist LU2.fna refs.msh` over all 8 references cited in the paper’s BLAST discussion.
7. **Whole-genome BLASTN.** `makeblastdb -dbtype nucl; blastn -perc_identity 70 -num_threads 16`; Python HSP-merge + length-weighted mean identity + query coverage.
8. **PlasmidFinder analog.** Cloned CGE `plasmidfinder_db` (488 replicon FASTA sequences); built BLAST DB; `blastn -perc_identity 95 -word_size 20`; applied PlasmidFinder default filter ($\geq 95\%$ identity AND $\geq 60\%$ subject coverage). **Zero replicons pass.**

9. **AMR gene scan.** `amrfinder -n LU2.fna -plus -threads 16 -o Escherichia`, AMRFinderPlus 3.12.8, DB 2024-07-22.1. 11 hits reported (Sec. 4.5).
10. **Metabolic-gene inventory.** Product-name regex over `/product=` tags in LU2.gb. All succinate-biosynthesis enzymes present (Sec. 4.4).
11. **LLM-judge scoring.** Full evidence dossier POSTed to `argo:claude-opus-4.6` at `http://127.0.0.1:44497` (T=0). Opus 4.7 and 4.8 were both returning HTTP 502 upstream-parser errors during this run; 4.6 was used as the highest working Opus sibling under the free-endpoint rule.

4 Results vs paper

4.1 Table 1 replication

Every quantitative row of the paper’s Table 1 for LU2 is reproduced exactly from CP035466.1 + barrnap:

	Metric	Paper	Our recount	Match
Genome size (bp)	5,062,651			5,062,651
Plasmids	0		0 (PlasmidFinder-analog)	
Contigs	1			1 (circular)
GC content (%)	55			55.00
Total genes	4,986			4,986
Protein-coding genes	4,741		4,741 (= 4,868 CDS – 127 pseudo)	
5S rRNA	8			8 (barrnap)
16S rRNA	7			7 (barrnap)
23S rRNA	7			7 (barrnap)
tRNA	86			86
ncRNA	10		9 ncRNA + 1 tmRNA = 10	
Pseudogenes	127			127
CRISPR arrays	0			0

4.2 Similar-Genome-Finder analog (Mash)

	Reference	Paper’s BLAST identity	Mash dist.	Rank match?	
<i>K. aerogenes</i>	AR0018 (CP024880.1)		99.57%	0.01176	#1
<i>K. aerogenes</i>	AR0007 (CP024883.1)		99.7%	0.01198	#2 (tie)
<i>K. aerogenes</i>	AR0161 (CP028951.1)		99.63%	0.01198	#2 (tie)
<i>K. aerogenes</i>	CAV1320 (CP011574.1)		99.25%	0.01208	#4
<i>K. aerogenes</i>	KCTC 2190 (CP002824.1)		99.44%	0.01390	#5
<i>E. cloacae</i>	ATCC 13047 (CP001918.1)		94.72%	0.17092	distant
<i>E. cloacae</i>	704SK10 (CP022148.1)		94.57%	0.19729	distant
<i>E. cloacae</i>	ECNIH7 (CP017990.1)		94.63%	0.19729	distant

Ranking is exact. Absolute Mash-ANI values are systematically lower than the paper’s megaBLAST identities because Mash counts shared 21-mers rather than aligned bases (this is a known Mash-vs-BLAST calibration offset, not a discordance).

4.3 Plasmid detection

PlasmidFinder-style BLAST vs 488 CGE replicon sequences at default thresholds (95% identity, $\geq 60\%$ subject coverage) returned **zero replicons**. All 7 raw sub-threshold hits are 20–25 bp fragments with e-value ≥ 1.9 (noise). **Paper’s “no plasmids” claim confirmed.**

4.4 Succinate biosynthesis genes

All enzymes of the paper’s proposed reductive-TCA + glyoxylate succinate route are present in the LU2 GenBank annotation:

Gene	Function	Present
<i>ppc</i>	PEP carboxylase (OAA replenishment)	
<i>mdh</i>	malate dehydrogenase (OAA $\rightarrow$ malate)	
<i>fumABC/D</i>	fumarase (malate $\leftrightarrow$ fumarate)	(4 hits)
<i>frdABCD</i>	fumarate reductase (fumarate $\rightarrow$ succinate)	(3 subunits)
<i>aceA</i>	isocitrate lyase (glyoxylate shunt)	
<i>aceB</i>	malate synthase A (glyoxylate shunt)	
<i>pykA/F</i>	pyruvate kinase	
<i>ppsA</i>	PEP synthase	

4.5 AMR profile (AMRFinderPlus 3.12.8, DB 2024-07-22.1)

11 hits, all intrinsic chromosomal loci (no horizontally-acquired transferable elements): *iroBCN* (salmochelins virulence), *uhpT*-E350Q (fosfomycin), *emrD* (efflux), *fieF* (metal), *fosA* (fosfomycin), ***ampC*** (*CMY-2/MIR/ACT/EC* family Class C β -lactamase, 99.21% identity), *kdeA* (efflux), *oqxA/oqxB* (multidrug efflux). AMRFinderPlus *confirms* the paper’s β -lactam finding and *extends* it with intrinsic quinolone (*oqxAB*) and fosfomycin (*fosA*, *uhpT*-E350Q) resistance signatures that ResFinder 3.1 (2020) would not have detected. Conversely, *sulI*, *uppP*, and *cat3* (reported by the paper) were not surfaced by AMRFinderPlus — most likely a DB/scope difference (AMRFinderPlus prioritises high-consequence acquired resistance over common Enterobacteriaceae chromosomal loci) rather than a contradiction.

4.6 CRISPR-Cas absence

Product-name grep for CRISPR|Cas1|Cas9|Csn2 across all 4,986 annotated genes returned **0 hits**. Consistent with the paper’s finding.

5 Per-claim what-worked / what-didn’t

C1 (chromosome + no plasmids) – strongly reproduced. Length 5,062,651 bp exact; circular topology confirmed from LOCUS line; PlasmidFinder-analog $\rightarrow$ 0 replicons pass.

C2 (GC 55%) – strongly reproduced. Computed 55.00% directly from the FASTA.

C3 (gene/RNA/pseudo counts) – strongly reproduced (exact on all 6 counts).

C4 (SPAdes hybrid assembly) – BLOCKED. Raw Illumina + ONT reads were never deposited to SRA. Reassembly impossible.

- C5 (comparative genomics) – strongly reproduced qualitatively.** Ranking of the 8 references matches the paper’s BLAST ordering exactly. Absolute identity values shift by aligner choice (~1–2 pp) but the *K. aerogenes*-vs-*E. cloacae* divide is preserved.
- C6 (succinate pathway) – strongly reproduced (annotation-level).** All 8 named enzymes present. Actual flux not measured (open question).
- C7 (AMR) – moderately reproduced.** β -lactam + efflux confirmed and extended; *sulI*/*uppP*/*cat3* not re-detected (DB delta).
- C8 (no CRISPR) – strongly reproduced.** 0 Cas-machinery genes in annotation.
- C9 (one intact prophage) – weakly checked.** Prophage machinery present in annotation, but PHASTER not re-run and specific boundary + RE-2010 match unverified.
- C10 (two bacteriocin clusters) – weakly checked.** Colicin annotations present, BAGEL4 not re-run.

6 Verdict

PARTIAL. The paper’s finished-genome science (structural + comparative + metabolic-annotation claims) is solidly reproducible on real deposited data with free open tools. The paper’s primary-sequencing-data science (hybrid SPAdes assembly, C4) is not reproducible because the raw reads were never deposited. Two web-only specialty analyses (PHASTER, BAGEL4) were plausibility-checked but not fully re-run. If the raw reads were deposited today, this replication would upgrade to REPLICATED within one additional wave.

7 Open Questions

- Q1. Aligner-choice contribution to the LU2-vs-KCTC ANI shift (paper 99.44%, ours 98.26%).**
Basis: we used `blastn -perc_identity 70` (permissive); the paper likely used web-BLAST megaBLAST defaults. Mash on the same pair gives ~98.61%. **Next steps:** run `pyani` (ANIm, ANIb) and FastANI on the same 8-reference set; compare all three aligners to isolate methodology-vs-biology; contact author for exact BLAST parameters if needed.
- Q2. Are the intrinsic *oqxAB* + *fosA* + *uhpT*-E350Q signals AMRFinderPlus surfaces in LU2 phen**
Basis: 11 loci found now vs ~5 in the paper’s 2020 ResFinder 3.1 scan. This is a DB delta, but silent DNA presence does not equal expressed resistance — and LU2 is being proposed as an industrial chassis where AMR-freedom matters. **Next steps:** MIC assays on LU2 for ciprofloxacin (*oqxAB*) and fosfomycin (*fosA*, *uhpT*); rerun current ResFinder 4.x to separate DB-vs-tool contributions.
- Q3. Is the reductive-TCA + glyoxylate route actually used for succinate biosynthesis *in vivo*, or is**
Basis: we confirmed every pathway gene is present, but presence $\neq$ flux. The paper never measured actual carbon flux. **Next steps:** ^{13}C -MFA on LU2 grown on U- ^{13}C -lactose whey permeate; if reductive branch is dominant, paper’s mechanism is supported; if not, mechanism needs revision.
- Q4. Has PHASTER’s ~4 $\times$ larger reference DB (2026 vs 2019) changed the identity, boundaries, or**
Basis: paper’s PHASTER call was made against the 2019 DB; the annotation clearly has

prophage machinery, but the specific boundaries and best-hit reference (*Salmonella phage RE-2010*) are 2019-DB-specific. **Next steps:** rerun PHASTER/PHASTEST; also run VirSorter2 and geNomad as independent second opinions.

Q5. Given LU2’s rumen ecology, is its mobilome burden (prophage + 47 GIs + many IS/tandem r

Basis: paper reports mobile-element counts for LU2 alone but never runs the same IslandViewer 4 + PHASTER + IS-scan pipeline on the 22 comparator *K. aerogenes* genomes. **Next steps:** run identical IslandViewer 4, PHASTER, and ISEScan pipelines on all 22+ complete *K. aerogenes* genomes at NCBI; compute per-Mb GI/IS/prophage burden; test for LU2 outlier status; correlate with each strain’s ecological source (clinical / environmental / rumen) to rigorously test the rumen-HGT-enrichment hypothesis.

8 Provenance

- LLM judge: `argo:claude-opus-4.6` at `http://127.0.0.1:44497/v1` (Argo proxy, free, key `stevens`). Opus 4.7/4.8 were both returning HTTP 502 during this run.
- All compute on `uicgpu` (8×A100).
- No paid endpoints used at any step.